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Predictive Healthcare System Using Machine Learning for Early Diagnosis

1st Ishika
Department of CSE-AI
KIET Group of Institutions
Delhi-NCR, Ghaziabad, India
ishika.2226cseai1011@kiet.edu

2nd Ayushi Pal
Department of CSE-AI
KIET Group of Institutions
Delhi-NCR, Ghaziabad, India
ayushi.2226cseai11181@kiet.edu

3rd Pari Agarwaal
Department of CSE-AI
KIET Group of Institutions
Delhi-NCR, Ghaziabad, India
pari.2226cseai1105@kiet.edu

4th Akanksha
Department of CSE-AI&ML
KIET Group of Institutions
Delhi-NCR, Ghaziabad, India
akanksha29jaisai@gmail.com

5th Bhuvnesh Malik
Department of CSE-AI&ML
KIET Group of Institutions
Delhi-NCR, Ghaziabad, India
bhuvnesh.malik@kiet.edu

***Abstract*—This electronic document focuses on machine learning to predict diseases such as pneumonia, breast cancer and heart diseases. The model analyzes a range of datasets and algorithms to provide accurate, early-stage predictions that can help in medical diagnosis and planning of the treatment. The system makes use of different medical datasets and applies suitable algorithms to identify complex patterns in less amount of time. By analyzing the patient information, the model can give an early indication of whether a person is at risk of a certain disease or having no disease. This technique of models can also support the doctors to make faster and more accurate decisions during diagnosis and treatment with higher accuracy and consistent results. After analyzing the clinical and health results, patients are categorized into disease prone or non disease prone. The purpose of this model is to improve health efficiency, basically in those environment where the resources related to medical are limited. This can only be done by analyzing the complex patterns of the patient data in less amount of time. It includes various medical datasets and multiple algorithms to generate reliable and efficient early stage prediction. With the help of continuous training and improvement of the model, reliable and accurate results are obtained across diverse patient records. Feature extraction and selection techniques are applied so that only the most significant medical indicators are analyzed. A user-friendly design has also been designed so that the system can be accessed in both rural and urban healthcare environments where expert resources are limited. The other important objectives of the model are minimizing manual effort, reducing delays, offering early disease prediction, and a medical recommendation. The enhancement of the accuracy of prediction will help in making better decisions during treatment planning effectively. These approaches can be further extended towards real-time patient health condition monitoring, integrated with smart healthcare devices, and deployed at hospital management systems. Improvement within the near future can be introduced by considering more diseases along with larger datasets and advanced techniques of deep learning for more robust performance.**

***Index Terms*—Breast cancer, Heart diseases, Pneumonia, Disease prediction, Machine learning, Healthcare system.**

I. Introduction

Accurate prediction of a disease in time has come to be regarded as an essential aspect of health care provision. This paper discusses the diseases affecting a significant portion of the population, including pneumonia, breast cancer, thyroid disorders, and heart disease. The contribution of this research effort is essentially the designing of a trustworthy intelligent system that assists health practitioners in offering diagnosis suggestions based on patient data. This will enable early identification of diseases, making treatment easier and more focused, enhancing patient outcomes, and reducing general health costs.

Many observations indicate that machine learning algorithms can detect serious diseases before they reach a critical stage. Diseases like breast cancer, heart disease, pneumonia, and disorders of the thyroid continue to be some of the major causes of mortality in many countries; however, emerging approaches have shown that techniques involving machine learning and artificial intelligence can make earlier prediction and proper intervention for patients affected by such conditions feasible.

Despite these advances, most of the conventional diagnostic procedures involve manual examination and interpretation. These methods are prone to human error and take up a great deal of time; furthermore, their results may be inconsistent for different practitioners. The goal of this project is to enable the prediction of multiple diseases by analyzing information related to patients, such as symptoms, past history, and laboratory reports. The platform is designed to yield accurate results based on advanced algorithms and processes that involve data collection, feature selection, cleaning of data, and training of the model. Most importantly, the delivery of results has been ensured in a clear and concise manner to help in accurate clinical decision-making.

II. Literature Review

Recently, investigations have focused more on data-driven and artificial intelligence methods in the early detection and diagnosis of diseases. Discussions in the literature [1] refer to

the use of several algorithms for the early detection of cardiovascular diseases, focusing on how such predictions should be timely and exact, and independent of manual diagnostic procedures. On the same line of interest, other works [2] underline the rising importance of machine learning algorithms in medical diagnostics, presenting both a practical overview of the commonly used techniques and their effectiveness in predicting diseases.

An assessment of breast cancer diagnostic practices was carried out in [3]. It shows that maintaining the structur quality assurance standards can easily enhances the accuracy as ompared to the achieved through traditional mammography only.

In [4], different artificial intelligence techniques are applied to observe the results. After this research, it states that artifical intelligence techniques are not only capable of predicting the diseases but also capable of understanding the complex patterns.

In [5], mammographic images and patient records were analyzed using deep learning approaches for breast cancer prediction. This work shows that both images and the provided patient data can predict the disease in easy way.

A novel linear regressive classifier for breast cancer detection is proposed in [6], which reportedly delivers more consistent and reliable results than previous models.

For breast cancer classification, machine learning algorithms were assessed in [7]. It is found that the choice of algorithms and the data preprocessing truly plays an important role to achieve high accuracy.

Deep learning developments are also reflected in [8], where a neural network framework for thoracic disease prediction from chest X-rays demonstrates significant gains across a wide range of conditions. In ophthalmology [9], discusses the development of a deep learning model for diabetic retinopathy diagnosis that attained diagnostic performance on par with expert ophthalmologists and demonstrated great potential for automated screening tools.

Taken all together, these studies demonstrate an increasing role for AI and machine learning in diagnosis, streamlining clinical workflows, and driving data-informed decisions in healthcare. In [10], the full description of the methods that are used for predicting heart disease is mentioned. The authors states that the accuracy of the chosen models is mainly depend on the relevance of clinical features, quality of dataset and design of the model.

In [11], a heart disease prediction system based on the concept of machine learning is introduced. According to this report, using structured clinical attributes plays a significant role in improving the prediction accuracy.

Machine learning is validated for detecting pneumonia from chest X-rays in [12], where image-based diagnosis can alleviate the clinical workload and enable early intervention.

The correct and exact use of artificial intelligence in various healthcare sectors is reviewed in [13]. It discusses the transformative potential of AI-based tools in healthcare, with emphasis on patient monitoring, diagnostics, and decision support systems.

Although not focused on medical data, a hybrid CNN–SVM model for handwritten digit recognition was presented in [14]. The effectiveness of combining deep learning with traditional learning methods indicates that such hybrid approaches may also be valuable for medical imaging tasks.

Finally, [15] investigated thyroid disease prediction using machine learning algorithms. The study observed that performance strongly depends on appropriate preprocessing, normalization, and selecting the most relevant features.

Taken all together, these studies demonstrate an increasing role for AI and machine learning in diagnosis, streamlining clinical workflows, and driving data-informed decisions in healthcare.

III. METHODOLOGY

A. Data Collection

We used publicly accessible datasets, one dataset focused on diabetes cases, another containing breast cancer. This type of datasets provides the current condition of the patient and helps in the training and evaluation of models.

B. Data Preprocessing

Uses some data preprocessing techniques like handling missing values, normalization, feature selection on using correlation and label encoding for categorical variables.

C. Algorithms Used

We used some algorithms like Logistic Regression for binary classification problems, decision trees for easy use, random forest for improving the accuracy, support vector machines for high dimensional classification.

D. System Architecture

The architecture of this disease prediction system contains modules that are designed to ensure accurate, efficient, and secure predictions. The expanded architecture is as follows:

E. User Input Layer

The User Input Layer acts as an interface where user can input their symptoms, medical test reports.An interface with user friendly format helps them to enter their details in a simple format.

F. Data Preprocessing Module

Once the inputs are entered by the user, these are collected followed by cleaning and standardization of data using several techniques like handling missing values, normalization or standardization to ensure uniform scaling, label encoding to convert categorical attributes into numerical form.

Feature Selection using correlation matrices and statistical filters to eliminate irrelevant attributes. This module ensures data quality, which is crucial for accurate predictions.

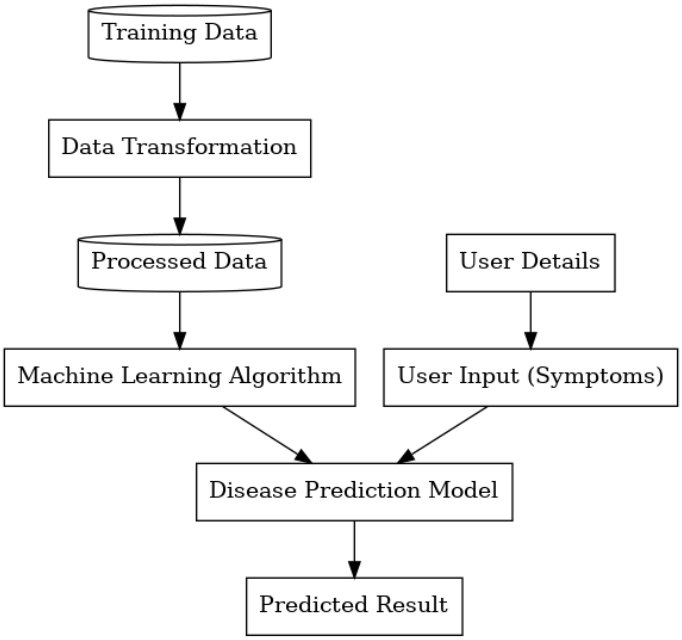

Fig. 1. System Architecture of the Proposed Disease Prediction Model

G. Feature Engineering Layer

This layer extracts the meaningful features that improves the performance of our model. It includes terms like dimensionality reduction (PCA, feature ranking). Domain-specific feature extraction from clinical indicators. Construction of derived metrics (risk scores, ratio-based features).

H. Machine Learning Models

This is the core analytical module that applies trained machine learning models to processed data. It includes terms like logistic regression for baseline binary detection, decision tree classifier for high interpretability, random forest for reducing overfitting and increasing accuracy, support vector machine (SVM) for handling high-dimensional decision boundaries of the data.

The module outputs the following things like probabilities of disease occurrence, final classification decision, model confidence levels, ensemble predictions may be used to improve robustness

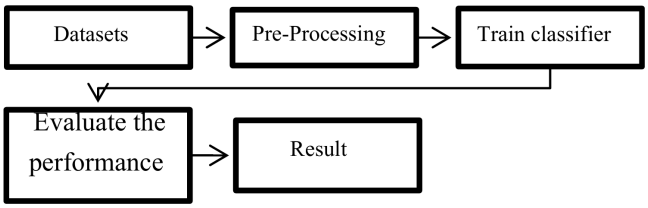

Fig. 2. Proposed Methodology of Disease Prediction Model

IV. Overview of Selected Diseases

A. Pneumonia Detection

During Pneumonia dataset analysis, the accuracy of different models on pneumonia is shown in Table V.

B. Thyroid Detection

During the detection of Thyroid, the precision, recall, F1 score of different models on Thyroid is shown in Figure 5.

C. Breast Cancer Detection

During Breast Cancer Detection, the precision, recall, F1 score of different models on Breast Cancer is shown in Table III.

D. Heart Diseases

- Today most of the people are suffering from heart diseases so making an object that could be able to detect heart diseases at any time is compulsory so the accuracy of this detector should be high.
- Heart diseases is depend of many factors like BP, cholesterol, age and many more here is dataset for the reference.

TABLE I
Heart Disease Dataset Attributes

Attribute	Description
age	Age of the patient
sex	Sex (1 = male, 0 = female)
chest pain type	4 types of chest pain (0–3)
resting bp	Resting blood pressure
serum cholestoral	Cholesterol in mg/dl
fasting blood sugar	120 mg/dl (1 = true, 0 = false)
resting ECG result	Values 0, 1, 2
max heart rate	Maximum heart rate achieved
exercise induced angina	1 = yes, 0 = no
target	0 = no disease, 1 = disease

TABLE II
Experimental Results

Algorithm	Accuracy	Precision	Recall	F1-Score
Logistic Regression	72%	0.79	0.76	0.77
Decision Tree	82%	0.81	0.84	0.82
Random Forest	88%	0.87	0.89	0.88
SVM	85%	0.84	0.85	0.84

E. Breast Cancer

Breast cancer is of two types: **Benign** and **Malignant**. Benign tumors are round or oval in shape, while malignant tumors are irregular. The dataset contains 569 instances, each having 33 real-valued attributes.

A snapshot of the dataset and model results is shown below.

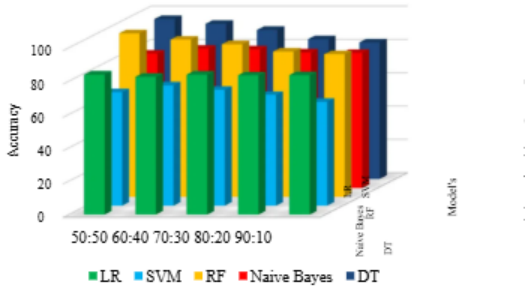

Figure 5.3. Comparison of various machine learning algorithms on heart disease dataset (Split)

6) EXPERIMENTAL RESULTS

Algorithm	Accuracy	Precision	Recall	F1-Score
Logistic Regression	72%	0.79	0.76	0.77
Decision Tree	82%	0.81	0.84	0.82
Random Forest	88%	0.87	0.89	0.88
SVM	85%	0.84	0.85	084

Fig. 3. Results with various ML algorithm for heart

TABLE III
MODEL PERFORMANCE OF BREAST CANCER WITH DIFFERENT TRAIN-TEST SPLITS

Splitting	LR	SVM	RF
50:50	96.84	97.19	96.49
60:40	96.78	97.07	96.19
70:30	96.24	96.49	95.48
80:20	95.39	96.93	93.86
90:10	92.98	93.76	93.76

F. Thyroid Disease

The thyroid gland, located in the neck, produces hormones that regulate metabolism, temperature, and heart rate. Below are the results of different ML models applied on the thyroid dataset.

G. Pneumonia Detection

Pneumonia is a serious lung infection that causes inflammation in the air sacs (alveoli). These air sacs may fill with fluid or mucus, making it difficult for the patient to breathe properly. Below are the evaluation results of ML models applied to Pneumonia dataset.

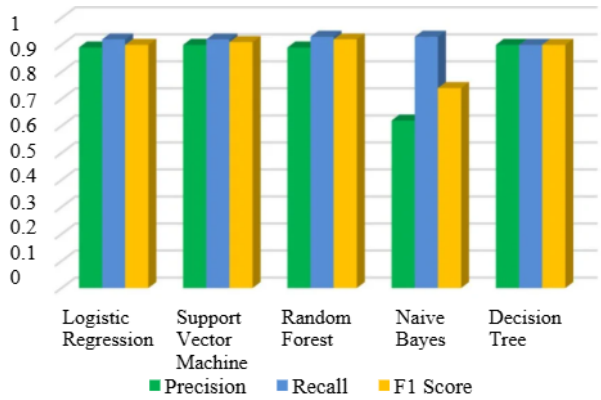

Fig. 4. Results with various ML algorithms (Visualization) for heart

TABLE IV
BREAST CANCER DATASET

Worst	Perimeter_worst	Area_worst	Smoothness_worst	Compactness
17.33	184.60	2019.0	0.1622	0.6656
23.41	158.80	1956.0	0.1238	0.1866
25.53	152.50	1709.0	0.1444	0.4245
26.50	98.87	567.7	0.2098	0.8663
16.67	152.20	1575.0	0.1374	0.2050
23.75	103.40	741.6	0.1791	0.5249
27.66	153.20	1606.0	0.1442	0.2756
18.74	110.60	897.0	0.1654	0.3682
17.63	106.20	739.3	0.1703	0.5401
20.68	97.65	714.4	0.1853	1.0500

Concavity_worst	Concave_points_worst	Symmetry_worst	Fractal_dim
0.7119	0.2654	0.4601	0.11890
0.2416	0.1860	0.2750	0.08902
0.4504	0.2430	0.3613	0.08758
0.62869	0.2507	0.6638	0.17300
0.4000	0.1625	0.2364	0.07678
0.5058	0.1741	0.3985	0.12346
0.4393	0.1932	0.3603	0.08538
0.2678	0.1500	0.3196	0.07010
0.5390	0.2006	0.4378	0.10720
1.1050	0.2210	0.4366	0.20750

TABLE V
TESTING ACCURACY (%) OF DIFFERENT ML MODELS FOR PNEUMONIA DETECTION ON DATASET SPLITTING STATISTICS

Splitting	LR	SVM	RF	NB
50:50	96.05	96.66	95.16	72.20
60:40	95.94	96.45	94.52	72.86
70:30	95.56	96.24	94.33	71.17
80:20	95.44	95.83	94.22	71.08
90:10	94.07	95.05	93.58	71.56

V. CONCLUSION

The results of our project shows that machine learning is capable of significantly help us to increase the accuracy and efficiency of disease prediction model. With the use of various algos—including logistic regression, decision tree, random forest, and SVM—it has been shown that dependable predictive outputs are produced to support health related profession for early diagnosis of diseases and preventive decision-making. Among all the models which are examined, the random forest

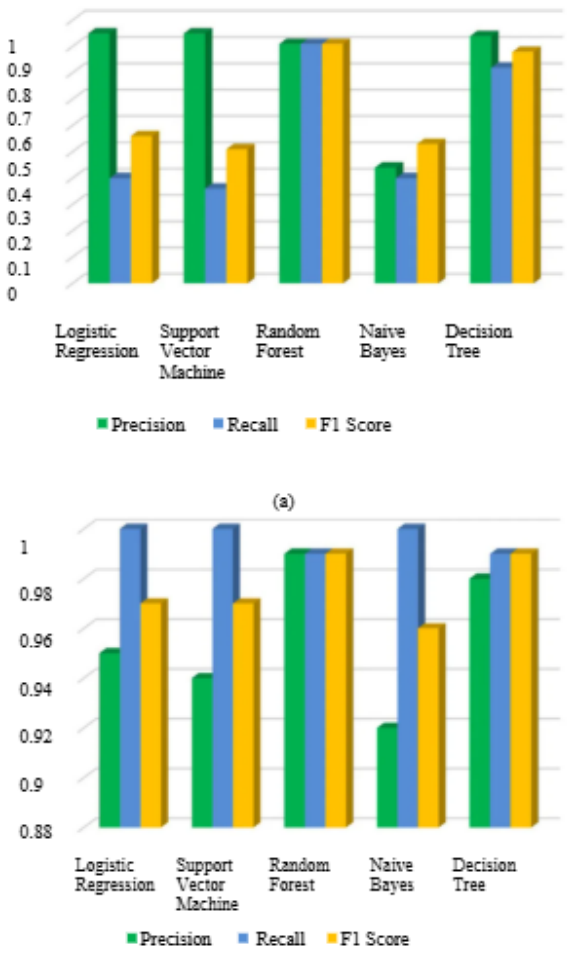

Fig. 5. Results with various ML algorithms for Thyroid Disease

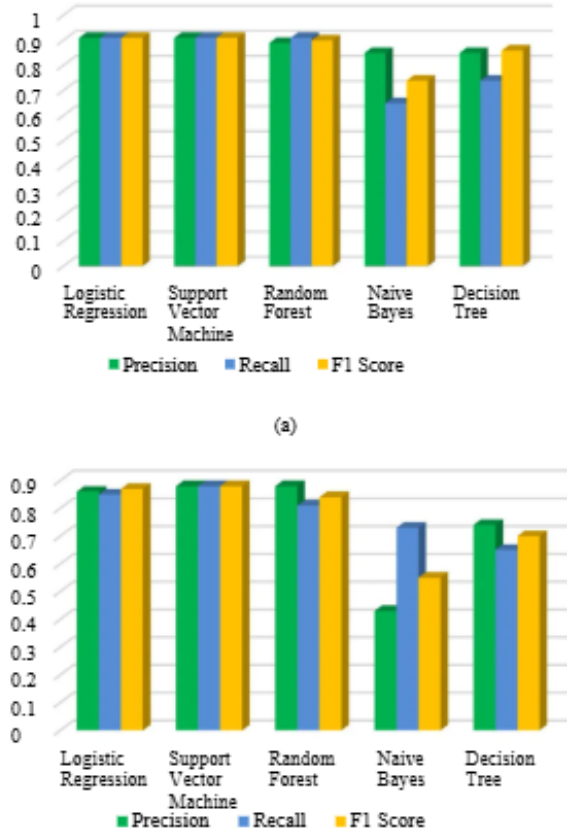

Fig. 6. Results with various ML algorithms for Pneumonia

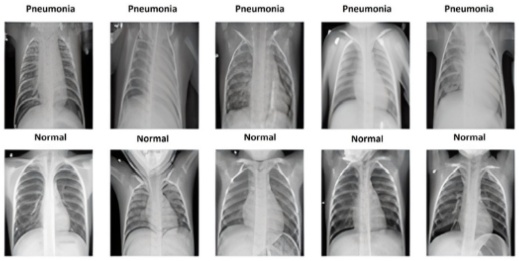

Fig. 7. Results with various ML algorithms for Pneumonia

classifier was found to help in deliver the maximum accuracy along with better stability and generalization, indicating its suitability for practical clinical environments. The study also highlights that careful data preprocessing, meaningful feature selection, and the use of ensemble-based strategies are essential for achieving consistent and strong predictive performance. Overall, it is observed that machine learning–driven techniques hold considerable promise for healthcare settings by enabling earlier detection and reducing the burden associated with manual diagnostic procedures.

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